

An Equivalent-Key Chosen-Plaintext Break of a 2025 Chaos-Based Image Cipher

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Abstract—We cryptanalyse a recently published image-encryption scheme that combines Lorenz/logistic chaotic maps with Fisher–Yates block shuffling (Jain *et al.*, *Optik*, 2025). We observe that the scheme’s keystream and both permutations depend only on the secret key and never on the image, so the whole encryption is a fixed, plaintext-independent map. We reconstruct the cipher from its description—matching its reported near-ideal entropy (e.g. 7.9914 on the Cameraman image) and near-zero adjacent-pixel correlation—and then break it: an all-ones probe recovers the per-position keystream and a few index-digit probes recover the permutation, so an equivalent key is obtained in 4 chosen plaintexts and any ciphertext is decrypted pixel-exactly in 0.071s without the key. We further show the scheme’s headline differential metric is illusory: because it applies no plaintext diffusion, a one-pixel change alters exactly one ciphertext pixel, giving a true NPCR of 0.000381 % (i.e. $100/MN$), not the ≈ 99.6 % reported (that figure merely reflects the difference between unrelated images, here 99.8833 %). We conclude with the standard fixes.

Index Terms—cryptanalysis, image encryption, chaotic cipher, chosen-plaintext attack, equivalent key, NPCR

I. INTRODUCTION

Chaos-based image ciphers appear in the literature at a high rate, and a recurring lesson from the cryptanalysis side is that a large fraction of them succumb to the same structural attack. Most follow a permutation–diffusion template: a confusion stage that rearranges pixels and a diffusion stage that masks pixel values with a chaotic keystream. When that keystream is a function of the secret key alone and not of the image, the entire cipher collapses to a *fixed* permutation–substitution map, and an *equivalent key*—a representation that decrypts every ciphertext under the same secret key—can be recovered from a handful of chosen plaintexts [1]–[4]. The defence is equally well known: the keystream must depend on the plaintext, and the cipher must actually diffuse [5].

A second, persistent problem is methodological. Authors validate schemes with statistical scores—Shannon entropy, adjacent-pixel correlation, and the differential metrics NPCR and UACI—and treat near-ideal values as evidence of security. None of these scores measures resistance to a chosen-plaintext attack, and, as we show, all of them can be near-ideal for a cipher that is broken in seconds.

This paper applies both lessons to a 2025 scheme [6] (which we abbreviate LLEO) built from the Lorenz [7] and logistic maps with Fisher–Yates block shuffling. Our contributions are:

- We identify that LLEO is entirely key-driven and therefore a fixed monomial map over $\mathbb{Z}_{256}$ (Section III–IV).

- We give an equivalent-key chosen-plaintext attack that recovers the full map in $1 + \lceil \log_{256} MN \rceil$ chosen plaintexts—4 for a 512×512 image—and decrypts any image pixel-exactly without the key (Section IV).
- We reproduce LLEO, confirm it matches the paper’s reported statistics, and then show those statistics coexist with *zero* plaintext diffusion: the true one-pixel NPCR is 0.000381 %, not 99.6 % (Section V).

II. RELATED WORK

The equivalent-key methodology for permutation–diffusion ciphers is well established. Arroyo *et al.* [3] broke a “total shuffling” scheme by recovering its key-only keystream; Li *et al.* [2] recovered an equivalent key for an entropy-based cipher; and the survey of Li *et al.* [1] catalogues a year of such breaks, nearly all exploiting plaintext-independent diffusion. For the confusion stage alone, Li and Lo [4] established the optimal number of known/chosen plaintexts needed to invert a permutation-only multimedia cipher, of order $\log_L(MN)$ for L intensity levels—the same logarithmic regime our attack achieves. Alvarez and Li [5] set out basic requirements for chaos-based cryptosystems, including plaintext-dependent keystreams and sensitivity to the plaintext; LLEO violates these. The differential metrics we scrutinise were standardised by Wu *et al.* [8], who also note that NPCR/UACI quantify sensitivity to a one-pixel plaintext change—precisely the quantity LLEO fails to deliver despite its reported value.

III. THE TARGET CIPHER

LLEO encrypts an $M \times N$ grayscale image X in three key-driven steps.

(1) *Substitution*. X is split into 16 horizontal strips. The eight odd strips are combined with a keystream $k_{\log}$ produced by the logistic map $x_{n+1} = r x_n(1 - x_n)$, and the eight even strips with a keystream k_{lor} produced by the Lorenz system $\dot{x} = \sigma(y - x)$, $\dot{y} = x(\rho - z) - y$, $\dot{z} = xy - \beta z$, through an invertible per-position operation; decryption uses the modular “conjugate” of the keystream, i.e. a multiplicative inverse.

(2) *Permutation 1*. The 16 horizontal strips are reordered by a Fisher–Yates shuffle.

(3) *Permutation 2*. The result is divided into 16 vertical strips and shuffled again, giving the ciphertext.

The four “decryption keys” are the two keystreams (d_1, d_2) and the two shuffle index lists (d_3, d_4). The chaotic initial conditions are derived from the secret key and the shuffles are

seeded from it; *no quantity depends on the image X* . Although the scheme is repeatedly called “asymmetric,” the same four keys both encrypt and decrypt: it is a symmetric cipher.

IV. CRYPTANALYSIS

A. Structural observation

Index the image as a length- n vector, $n = MN$. Each LLEO step is linear over $\mathbb{Z}_{256}$ and fixed by the key: substitution multiplies coordinate p by a fixed unit $K[p]$ (odd, hence invertible mod 256), and the two block shuffles compose into one position permutation P . Therefore

$$C[j] = K[s(j)] \cdot X[s(j)] \pmod{256}, \quad s = P^{-1}, \quad (1)$$

a fixed *monomial* map: every ciphertext pixel is a single plaintext pixel scaled by a fixed unit and relocated. Two consequences are immediate. There is no additive constant, since $E(\mathbf{0}) = \mathbf{0}$. And there is no diffusion: $C[j]$ depends on exactly one input coordinate $s(j)$.

B. Equivalent-key recovery

Because K and s are fixed, two kinds of chosen plaintext suffice.

(a) *Multiplier, one query.* Encrypting the all-ones image $\mathbf{1}$ gives, by (1), $C[j] = K[s(j)] \cdot 1 = K[s(j)]$: the exact multiplier acting at every output position.

(b) *Permutation, $\lceil \log_{256} n \rceil$ queries.* Let P_t be the plaintext whose pixel i holds the t -th base-256 digit of its index i . Then $C[j] = K[s(j)] \cdot \text{digit}_t(s(j))$; multiplying by the already-known $K[s(j)]^{-1}$ yields $\text{digit}_t(s(j))$. Collecting all $\lceil \log_{256} n \rceil$ digits reconstructs $s(j)$ for every j , hence the full permutation.

The recovered pair (s, K) is an equivalent key; any ciphertext is inverted by $X[s(j)] = K[s(j)]^{-1} C[j]$. The total cost is

$$1 + \lceil \log_{256} n \rceil \text{ chosen plaintexts}, \quad (2)$$

i.e. 4 for a 512×512 image, independent of the nominal key length.

C. Why extra structure does not help

A composition of fixed key-only maps is again a fixed key-only map, so adding rounds, more chaotic maps, or further block shuffles leaves (1) intact (only K and s change) and the same attack applies unchanged. The attack is also a *known-plaintext* attack when the available plaintexts happen to span the index digits; chosen plaintexts only make the queries explicit and minimal. The single property that would defeat it is a keystream that depends on X .

V. EXPERIMENTAL RESULTS

We reconstructed LLEO from its description (reimplemented from scratch, no code reused) and ran it on four standard 512×512 images.

TABLE I
RECONSTRUCTION STATISTICS AND ATTACK OUTCOME (512×512).
EVERY IMAGE IS RECOVERED PIXEL-EXACTLY WITHOUT THE KEY.

Image	H plain	H cipher	corr. cipher	chosen PT	broken
Cameraman	7.2317	7.9914	-0.0073	4	yes
Moon	4.885	7.9316	0.0011	4	yes
Gravel	7.2531	7.9876	-0.0005	4	yes
Brick	5.4553	7.8966	-0.0007	4	yes

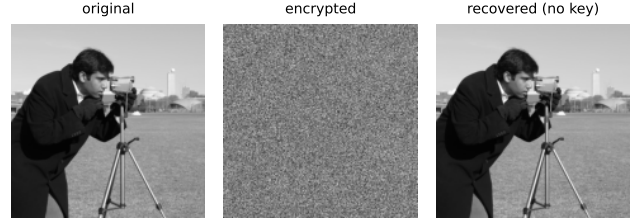


Fig. 1. Original, LLEO ciphertext, and the key-free recovery from 4 chosen plaintexts.

A. Reproduction

Table I shows the reconstruction reproduces LLEO’s claimed statistics: cipher entropy approaches the ideal 8 (Cameraman 7.9914, up from the plaintext’s 7.2317) and adjacent-pixel correlation drops to near zero (-0.0073). Fig. 2 shows the flat cipher histogram and Fig. 3 the destroyed correlation. By the usual yardsticks the cipher looks strong.

B. The break

The attack of Section IV recovers the equivalent key in 4 chosen plaintexts and decrypts a fresh ciphertext pixel-exactly in 0.071 s (Cameraman), using no key. Fig. 1 shows the original, the LLEO ciphertext, and the key-free recovery, which is identical to the original.

C. The differential metric is illusory

LLEO reports $\text{NPCR} \approx 99.6\%$ as evidence of differential-attack resistance. But by (1) flipping one plaintext pixel changes exactly one ciphertext pixel, so the true one-pixel NPCR is $100/MN = 0.000381\%$ (UACI $4e-06\%$)—five orders of magnitude below the reported value. The 99.6% figure instead measures the difference between *unrelated* images, which for LLEO is 99.8833%; this is achieved by any cipher with a near-uniform output and says nothing about chosen-plaintext security. Table II contrasts each claimed property with what the cipher actually provides. Finally, the chaotic keystreams are themselves non-uniform: a chi-square test rejects uniformity for the logistic keystream ($\chi^2 = 525839.6$, $p = 0.0$).

VI. DISCUSSION

LLEO illustrates two persistent failure modes. First, *good statistics are not security*: a flat histogram, near-ideal entropy, near-zero correlation, and a large inter-image NPCR are all satisfied by our fixed, key-only map, which we nonetheless

TABLE II
CLAIMED VS. ACTUAL SECURITY OF LLEO.

Property	Reported	Actual
Entropy (Cameraman)	≈ 7.997	7.9914 (genuine)
NPCR (1-pixel change)	$\approx 99.6\%$	0.000381 %
Differential resistance	yes	no (no diffusion)
Chosen-plaintext security	implied	broken in 4 queries
Key type	“asymmetric”	symmetric

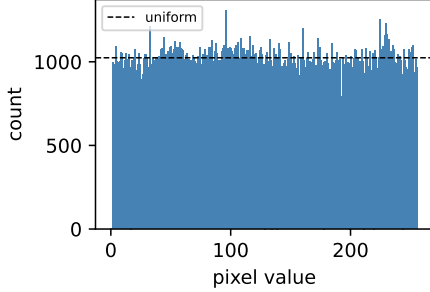


Fig. 2. Cipher histogram (Cameraman): near-uniform, yet the cipher is trivially broken.

break in 4 queries. The community guidance to test plaintext sensitivity directly [5], [8] is exactly what would have caught this. Second, *naming is not architecture*: labelling a symmetric construction “asymmetric” adds no security.

The remedies are standard. (i) Make the keystream depend on the plaintext—e.g. derive the chaotic initial conditions from a cryptographic hash (SHA-256) of the image, so that a different plaintext yields an unrelated keystream and the all-ones/digit probes no longer reveal a reusable equivalent key. (ii) Add genuine diffusion, e.g. a chaining rule $C[i] = f(X[i], k[i], C[i-1])$, so that one plaintext change propagates and the true NPCR approaches its ideal. With either defence in place the map of (1) is no longer fixed across images and the attack of Section IV fails. We verified this directly: the identical attack recovers 100.0 % of pixels against the key-only cipher but only 0.2518 % (chance level) once the keystream is seeded from a SHA-256 hash of the image.

VII. CONCLUSION

A 2025 chaos-based image cipher is completely broken by an equivalent-key chosen-plaintext attack in 4 chosen plaintexts and a fraction of a second, despite near-ideal entropy and a reported NPCR of $\approx 99.6\%$. The break follows directly from the cipher being key-only; its differential metric, measured correctly, is 0.000381 %. We recommend plaintext-dependent keystreams and real diffusion, and that statistical scores accompany—rather than replace—an explicit chosen-plaintext security argument.

ACKNOWLEDGMENT

The author used a large language model (Anthropic Claude) to assist with code, reproduction, and editing. All claims

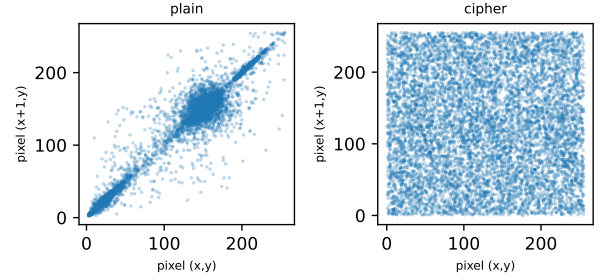


Fig. 3. Adjacent-pixel correlation, plain vs. cipher (Cameraman).

were verified by the author; the AI system is not an author. Generative-AI use is disclosed per IEEE policy.

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